BT

**BREATHTWIN**

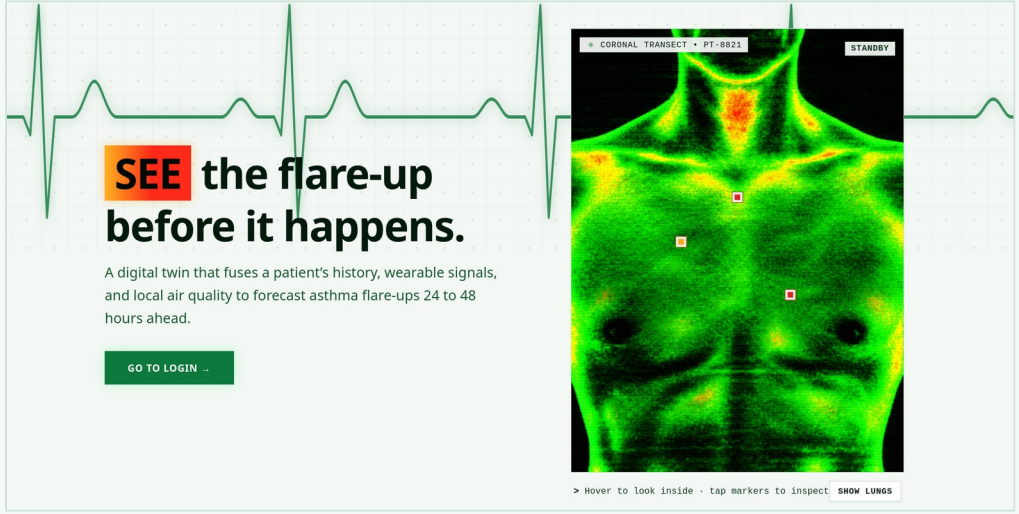
MULTIMODAL ASTHMA DIGITAL TWIN

PROJECT PRESENTATION

# BreathTwin

## A multimodal digital twin for forecasting asthma flare-up risk 24–48 hours ahead

Connect clinical history, wearable signals and outdoor air quality. Estimate a hidden airway state, forecast risk, and let clinicians explore “what-if” scenarios.




DOCTOR + PATIENT PORTALS

LNMIIT × NSUT

**Aarush Bhattacharjee** · **Team lead & full-stack / backend**

**Rakshit Gupta** · **Machine learning**

Open source • MIT License • Synthetic patient cohort

PROJECT REPOSITORY · github.com/Frisbee148/DigitalTwin

THE NEED

# The warning signals are spread across different places

A clinician may see the pieces at different times: a baseline in the record, a change in wearable readings, and a change in outdoor air quality.

01

## Clinical history

**What is normal for this patient?**

- Spirometry and lung function
- Controller regimen and adherence
- Previous flare-ups and care history

02

## Wearable signals

**What is changing right now?**

- Overnight oxygen and heart rate
- Resting respiratory rate and HRV
- Sleep, symptoms and rescue use

03

## Outdoor conditions

**What is the patient exposed to?**

- PM2.5 and PM10 concentrations
- Air Quality Index (AQI)
- Temperature and humidity

**BreathTwin's goal** Bring these signals together and give the care team time to review a risk estimate before an emergency becomes the first clear signal.

THE APPROACH

# One connected workflow, from signals to a useful next step

BreathTwin combines learned predictions with a mechanistic airway simulation, then presents the result through doctor and patient interfaces.

1

### COLLECT

Clinical baseline, 20-channel wearable window, and local air-quality data.

2

### ESTIMATE

A modelled airway-inflammation state that changes with exposure and controller adherence.

3

### FORECAST

24-hour and 48-hour flare-up risk, plus a severity estimate.

4

### EXPLORE

Compare simulated outcomes under different prevention scenarios.

5

### COMMUNICATE

Doctor triage, explainable drivers, and plain-language patient guidance.

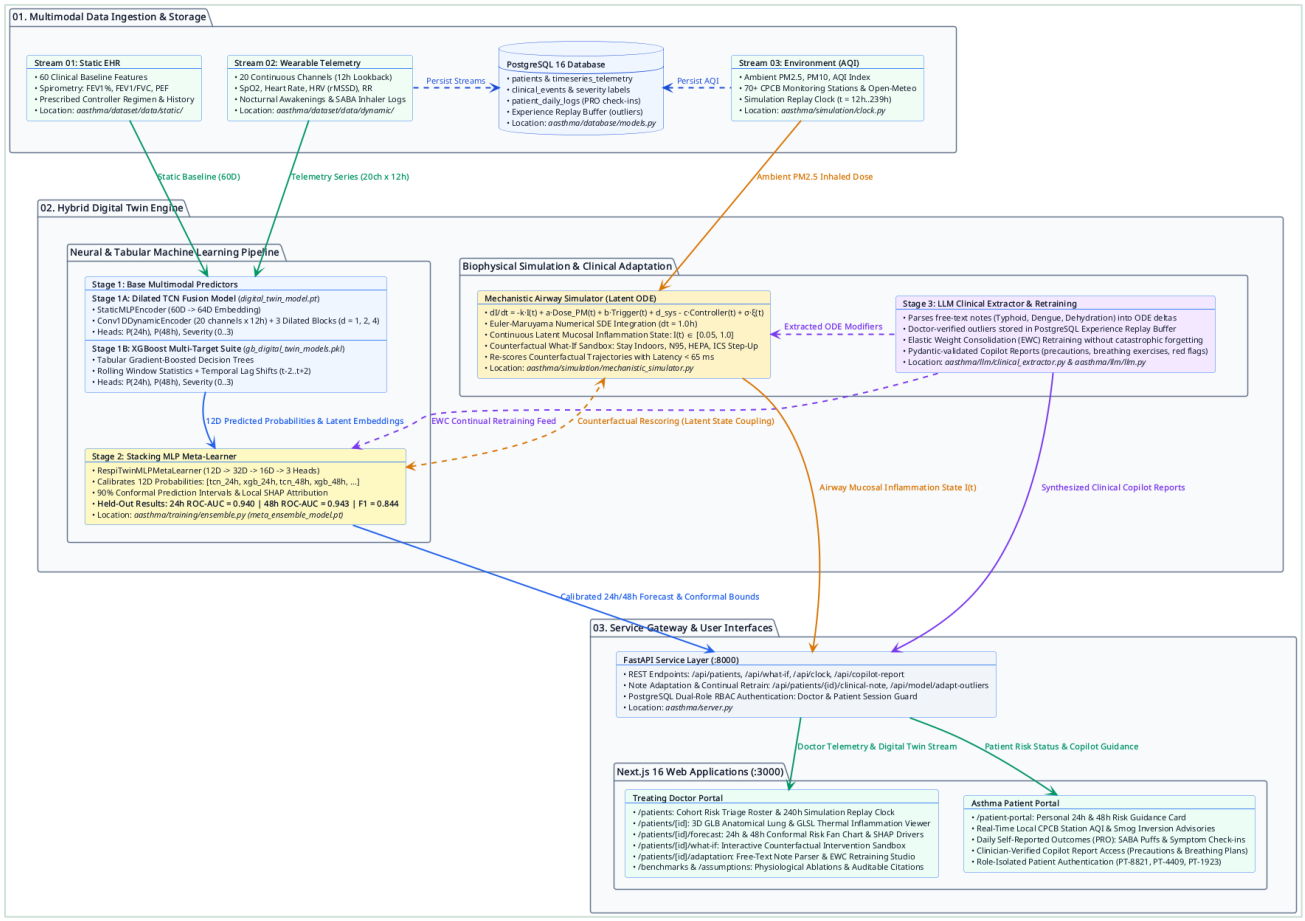
**What makes it a digital twin?** It is not only a risk score. The system maintains a modelled airway state, updates it as inputs change, and re-runs the forecast when a clinician tests a scenario.

**Clinical decision support only — not a diagnostic or prescribing system.**

SYSTEM ARCHITECTURE

# Three layers keep the project understandable

The diagram shows how incoming data feeds the hybrid twin engine, then reaches the application interfaces through the API layer.




**01** **Input & storage**

Static health records, wearable telemetry, environmental readings and PostgreSQL persistence.

**02** **Hybrid twin engine**

Dilated TCN + XGBoost + stacking MLP, supported by a mechanistic airway ODE and note adaptation.

**03** **Services & interfaces**

FastAPI endpoints serve doctor triage, forecasts, what-if tools and the patient companion.

*Architecture image supplied with the project*

DATA INGESTION

# Three streams describe the patient and their environment

Inputs are aligned into a common view; the clinical history and hourly telemetry provide the patient context while air quality describes an external exposure.

STREAM 01

## Static clinical baseline

**60 features**

Patient history and baseline status

- Age, sex, BMI and smoking history
- FEV1, FEV1/FVC and peak flow
- Controller regimen and adherence
- Past flare-ups and urgent care

STREAM 02

## Wearable telemetry

**20 channels × 12 h**

A rolling window of recent signals

- SpO2, heart rate, respiratory rate, HRV
- Skin temperature and activity
- Sleep and nocturnal awakenings
- Symptoms and rescue inhaler use

STREAM 03

## Environmental context

**PM2.5 / PM10 / AQI**

Local air and weather conditions

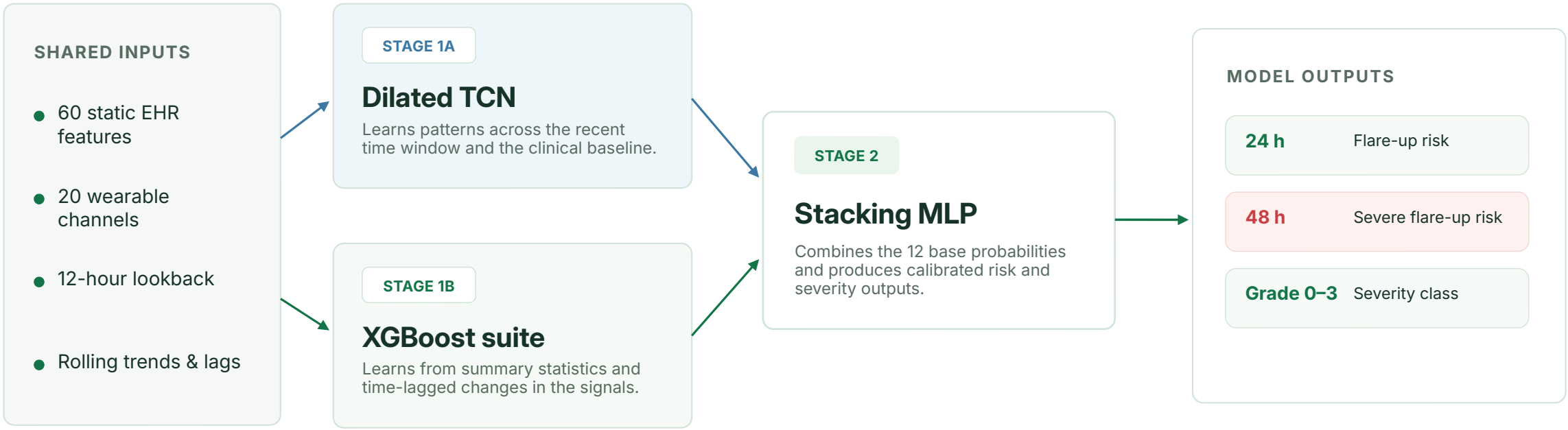
- Particulate matter levels
- AQI from monitoring stations
- Temperature and humidity
- Live feed or replayed conditions

**POSTGRESQL 16** Stores patient records, time-series telemetry, clinical events, daily logs and user accounts.

PREDICTIVE PIPELINE

# Two base models learn different patterns; a third combines their outputs

The ensemble uses recent time-series behaviour and engineered tabular features, then stacks their probabilities into a single multi-horizon estimate.

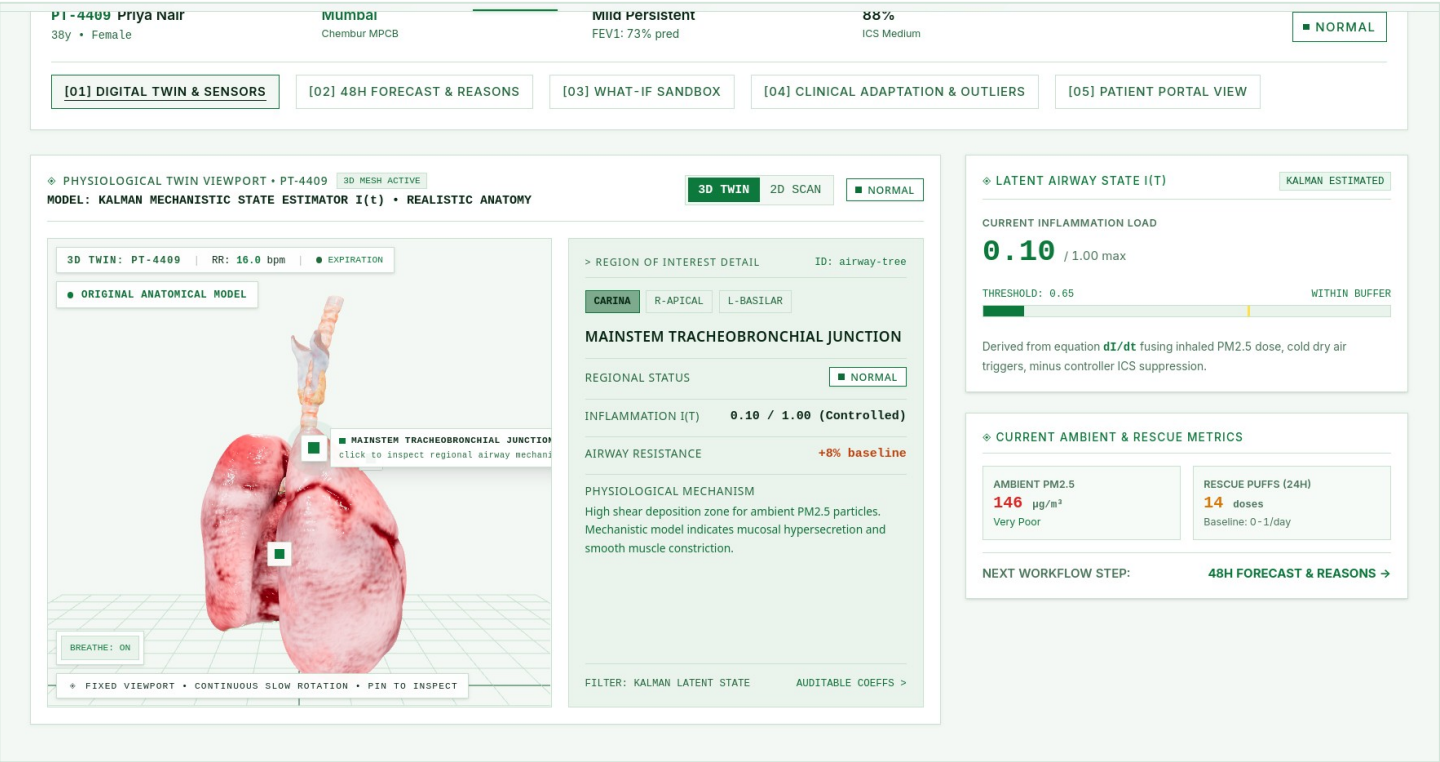



**MECHANISTIC CONTEXT** A separate airway ODE estimates the latent inflammation state and supports counterfactual re-scoring alongside the learned ensemble.

MECHANISTIC SIMULATION

# The twin tracks a modelled state the wearable cannot measure directly

A differential-equation simulator updates latent airway inflammation using exposure, symptoms, systemic triggers and controller effects.




## LATENT STATE I(t)

A continuously updated estimate of airway inflammation, bounded in the project model from 0.05 to 1.0.

**PROJECT THRESHOLD: 0.65**

## HOW THE STATE CHANGES

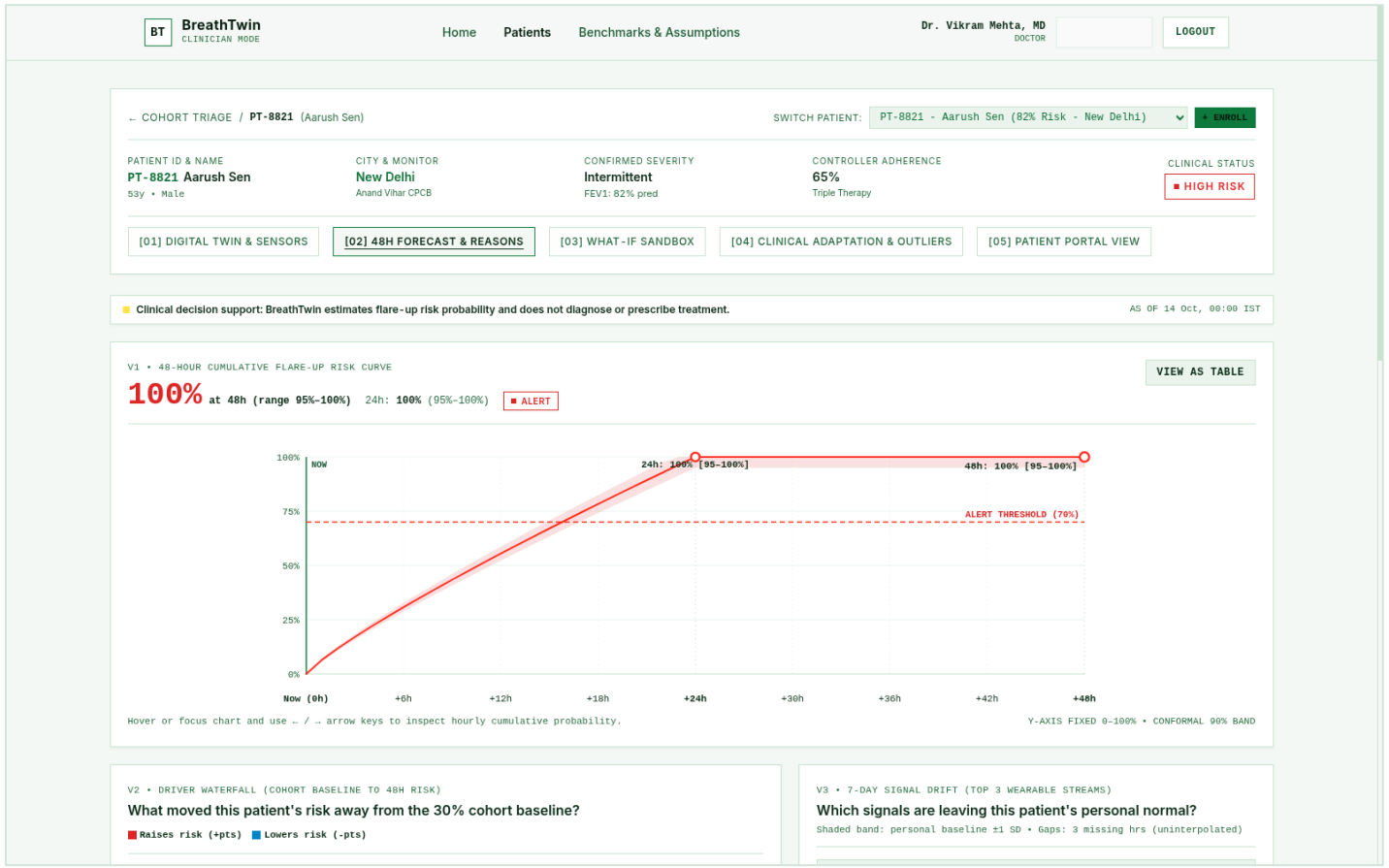
| | |
|---|---|
| **Recovery** | tends to lower inflammation |
| **PM2.5 dose** | can raise the modelled load |
| **Triggers / illness** | add extra perturbations |
| **Controller effect** | models anti-inflammatory influence |

*This is a simulated state, not a direct inflammation measurement or a clinically established universal threshold.*

FORECAST AND EXPLANATION

# Show the risk trajectory and the signals behind it

BreathTwin presents a forecast as a timeline with uncertainty and local driver attribution, rather than only a single risk number.




**01**

## Two forecast horizons

Separate estimates for 24-hour flare-up risk and 48-hour severe exacerbation risk.

**02**

## Uncertainty is visible

The project uses 90% conformal prediction intervals to show a range around the forecast.

**03**

## Risk drivers are inspectable

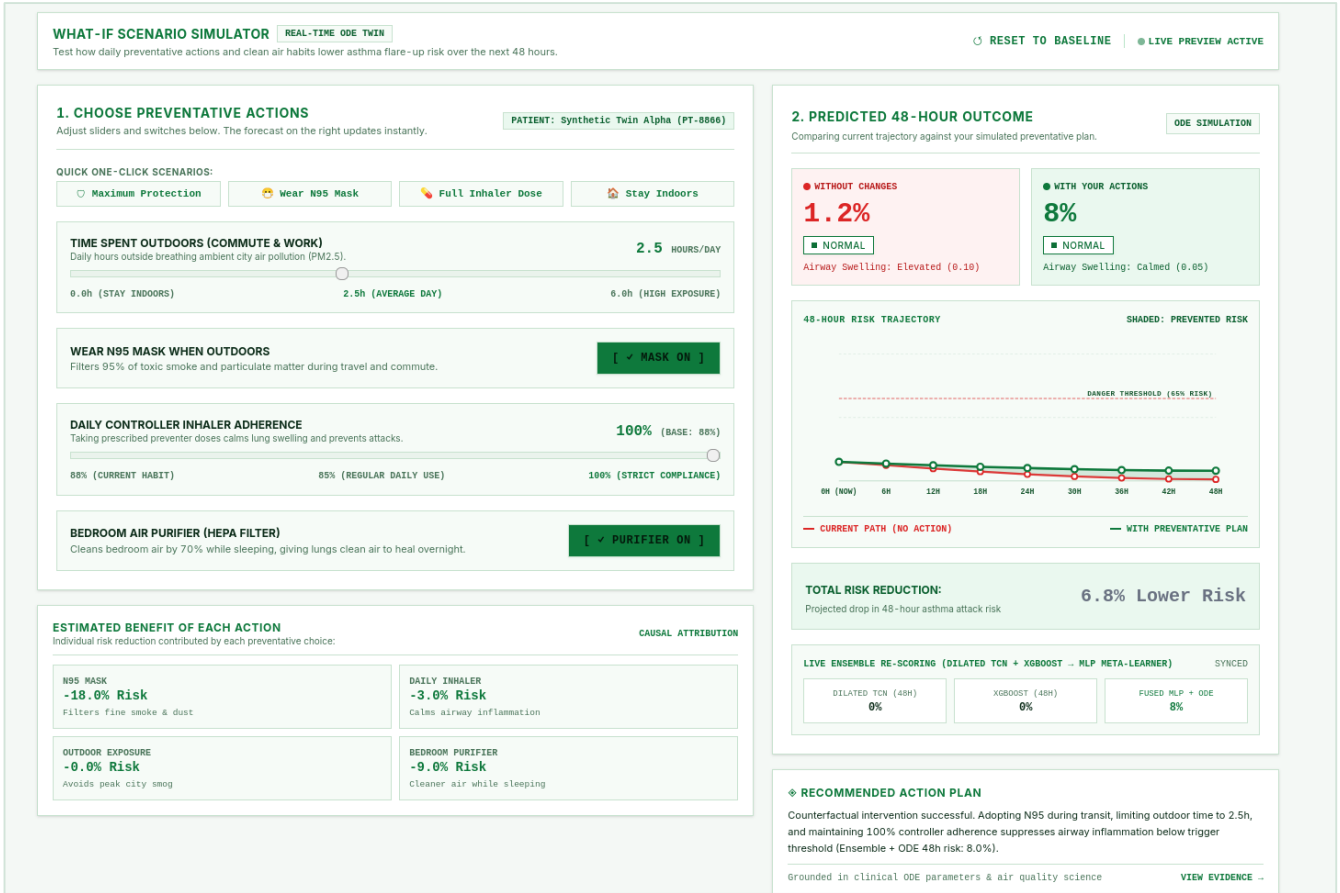
A local SHAP view ranks contributing factors such as rescue use, oxygen changes and particulate exposure.

**Conformal coverage must be validated on real-world data.**

INTERVENTION SANDBOX

# Ask “what if?” before comparing a prevention plan

The sandbox re-runs the model under changed assumptions and displays how the simulated trajectory differs from the current path.




## SCENARIOS IN THE PROJECT

| | |
|---|---|
| **Outdoor time** | Reduce the hours spent in polluted outdoor air. |
| **N95 mask** | Model filtration during commute and outdoor exposure. |
| **HEPA purifier** | Model lower indoor particulate exposure. |
| **Controller adherence** | Explore adherence assumptions within the care plan. |

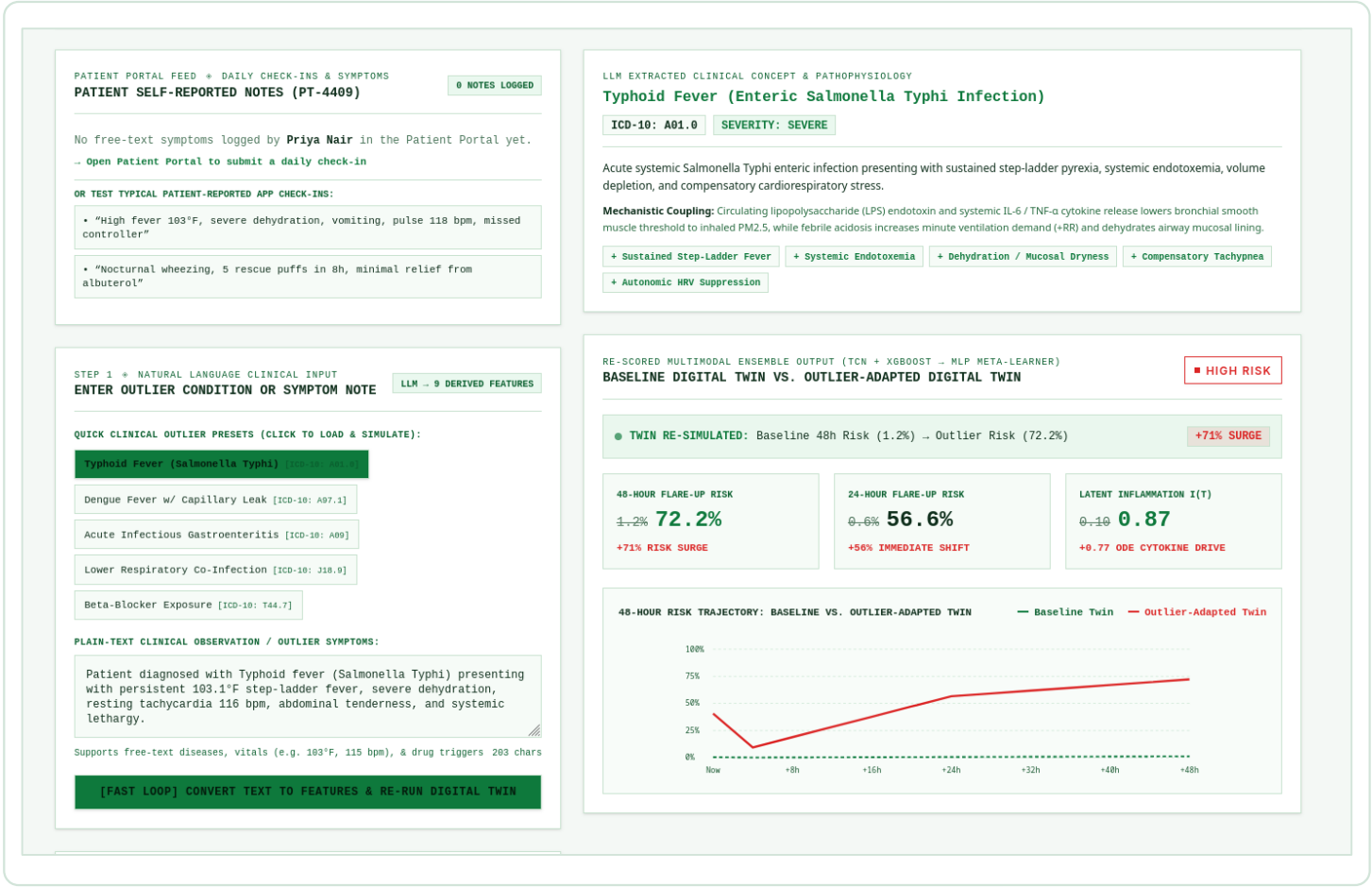
### Important boundary

These are modelled counterfactuals, not proof that an intervention will produce the same effect in a real patient. Medication changes remain clinician-led.

CLINICAL NOTE ADAPTATION

# Add context when the sensor stream is not the whole story

The prototype translates a clinician's free-text note into structured features, then re-scores the twin with those additional perturbations.




### 1 Enter a clinical note

For example, a concurrent infection, fever, dehydration or a medication change.

### 2 Extract structured features

The LLM / ontology pipeline maps text to condition labels, vital-sign changes and ODE modifiers.

### 3 Re-simulate and review

The twin updates its trajectory; a clinician can inspect the derived features and result.

### 4 Commit and adapt

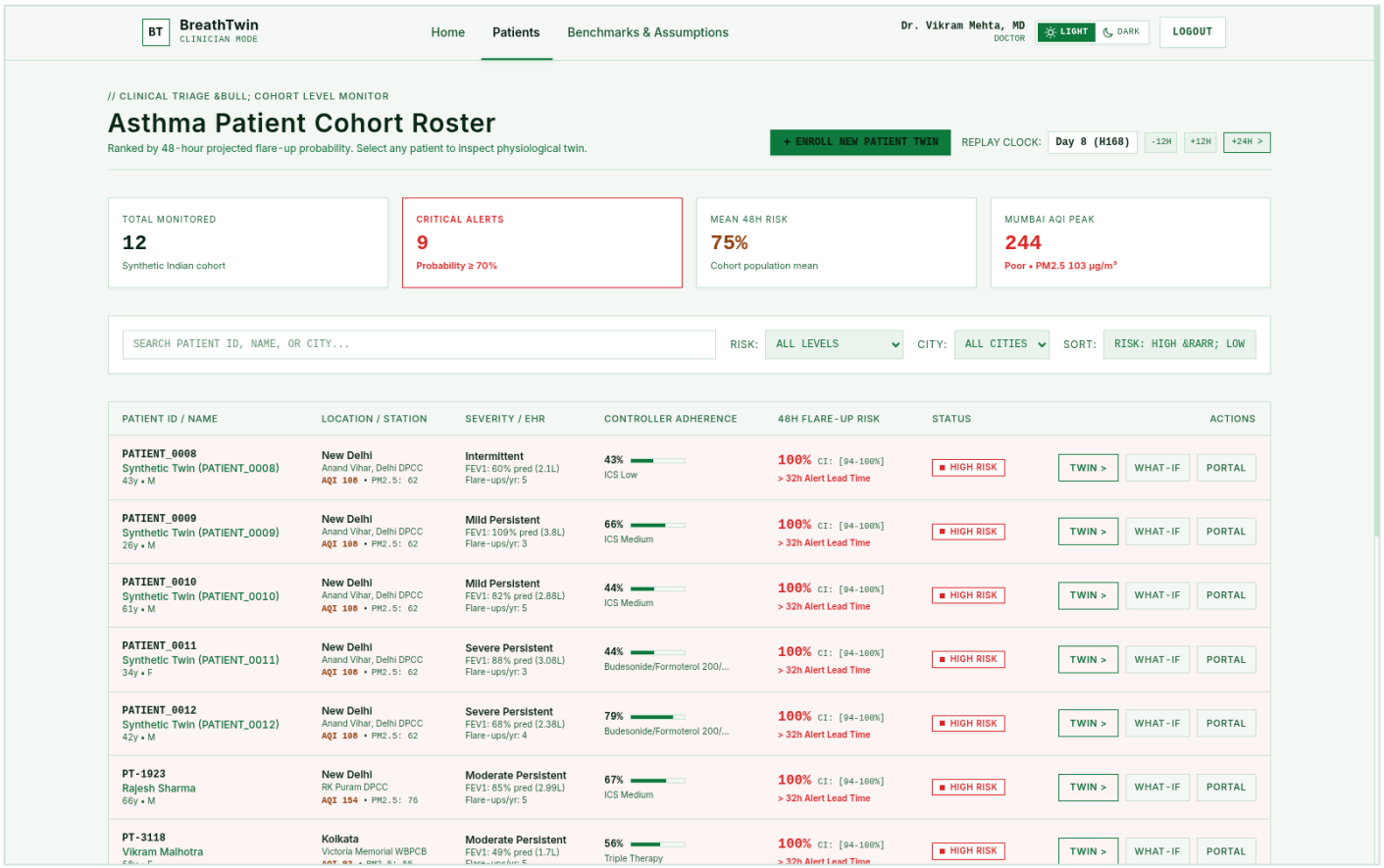
Verified outlier cases enter the replay buffer; EWC regularization aims to reduce forgetting during retraining.

*A clinician review and commit step is part of the stated workflow; extracted values should not be treated as verified facts automatically.*

DOCTOR WORKFLOW

# Start with the cohort, then inspect an individual patient

The clinician interface is designed to help a team prioritize attention, open an individual twin, and review forecasts in context.




*Illustrative dashboard values are from the supplied project screenshots and synthetic cohort.*

## COHORT VIEW

- Rank the patient roster by projected 48-hour risk.
- Filter by city, risk level and available patient context.
- Move from the roster to a 3D twin, forecast, what-if sandbox or note adaptation view.

## INDIVIDUAL VIEW

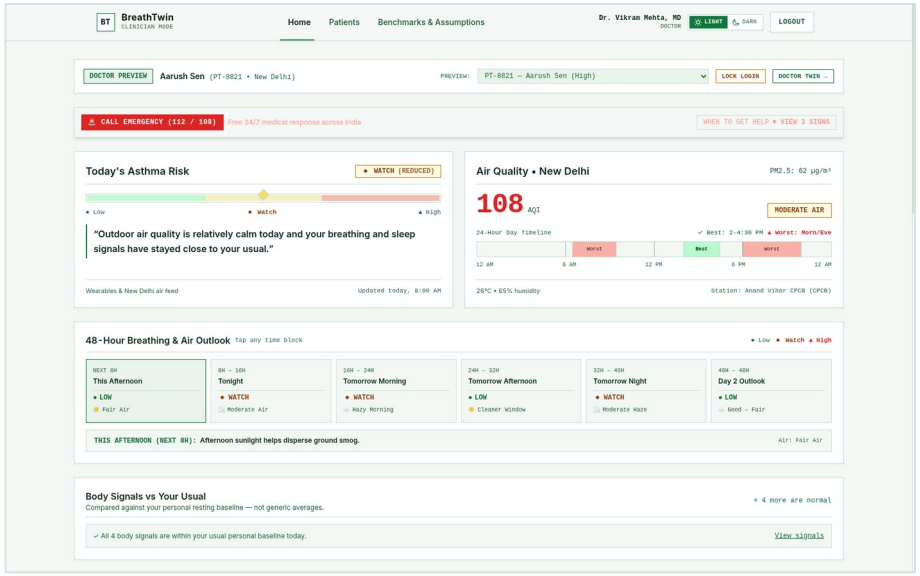
- Compare recent signals with a personal baseline.
- Review the drivers and projected trajectory before discussing a care plan.

PATIENT COMPANION

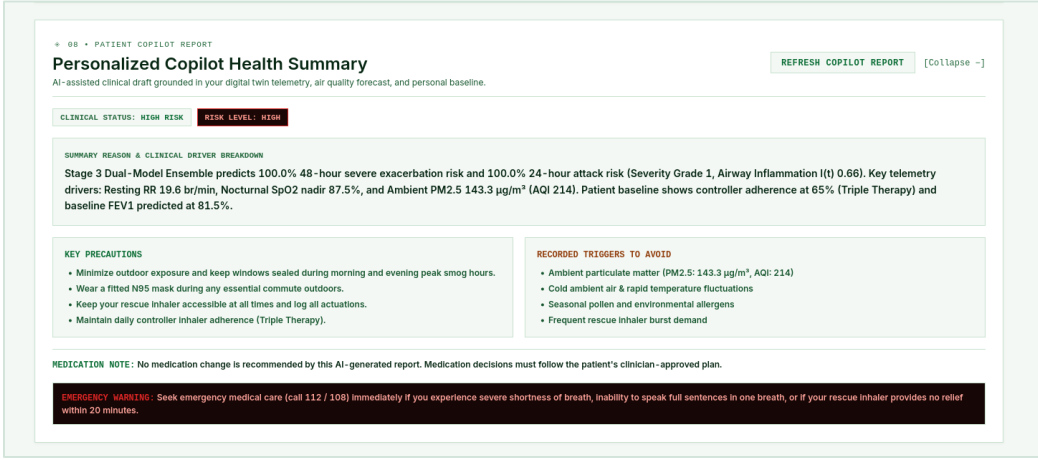
# Turn model output into guidance people can understand

The patient portal avoids technical model language and brings risk status, local air outlook, personal signals and daily check-ins into one view.

## 01 / DAILY STATUS & AIR OUTLOOK




## 02 / COPILOT HEALTH SUMMARY




### Plain-language risk

A simple status and a short explanation of likely drivers.

### Daily self-report

Logs symptoms, rescue use and controller adherence.

### Safety guardrails

No autonomous prescribing; emergency red flags remain explicit.

REPORTED EVALUATION

# Strong separation on the project's synthetic held-out cohort

The repository reports patient-level splits with 800 training, 100 validation and 100 held-out test patients (random seed 42).

**0.9399**
**24-hour ROC-AUC**
BreathTwin ensemble

**0.9428**
**48-hour ROC-AUC**
BreathTwin ensemble

**0.8435**
**Severity weighted F1**
4 severity grades

**< 65 ms**
**Counterfactual re-score**
Reported simulation latency

## ROC-AUC COMPARISON

| Model | 24 h ROC-AUC | 48 h ROC-AUC | Mean warning lead time |
|---|---|---|---|
| **BreathTwin (TCN + XGBoost + MLP + ODE)** | **0.9399** | **0.9428** | **42.4 ± 4.1 h** |
| LSTM baseline | 0.7850 | 0.7790 | 22.1 ± 5.8 h |
| XGBoost-only baseline | 0.7410 | 0.7320 | 18.6 ± 6.2 h |

**Interpretation limit** These are project-reported results on synthetic data. They do not establish performance, clinical utility or safety on real patients.

*Source: BreathTwin README, §7.6 (held-out test metrics). Values rounded only where shown in stat cards.*

SAFETY AND SCOPE

# Built as decision support, with important limits made explicit

The prototype includes guardrails, but its forecasts and simulated intervention effects require additional validation before clinical use.

## DESIGN SAFEGUARDS

### Synthetic patient records

The repository states that patient records are synthetic; do not interpret the screenshots as real patient cases.

### Clinician stays in control

BreathTwin is framed as decision support, not autonomous diagnosis or prescribing.

### Role separation and logs

Doctor and patient access are separated through role-based access and session controls; events are logged.

### Non-prescriptive copilot

Patient-facing output reinforces the clinician-approved plan and includes emergency red flags.

## WHAT IS NOT YET PROVEN

### Real-world prediction accuracy

Synthetic test metrics do not show how the model performs in clinical populations.

### Calibrated uncertainty in practice

Prediction intervals need coverage and calibration checks on representative real-world data.

### Causal benefit of actions

A simulated risk decrease is not proof an intervention will reduce a person's risk by the same amount.

### Clinical workflow readiness

Prospective evaluation, governance, privacy review and clinician assessment are still required.

PROJECT HANDOFF

# An open-source prototype built by an inter-college team

BreathTwin brings the model pipeline, physiological simulator, application interfaces and reproducible local setup into one repository.

## TECHNICAL STACK

| | |
|---|---|
| **Frontend** | Next.js 16 · React 19 · TypeScript · Tailwind CSS |
| **3D experience** | Three.js · WebGL · GLSL shaders |
| **API and services** | Python · FastAPI · Pydantic · Uvicorn |
| **Models** | PyTorch · Dilated TCN · XGBoost · Stacking MLP |
| **Simulation and data** | Mechanistic ODE · PostgreSQL 16 · SQLAlchemy |
| **Delivery and tests** | Docker Compose · Pytest · run.sh |

## TEAM & CONTRIBUTIONS

### Aarush Bhattacharjee

LNMIIT · Team lead, system architecture, backend / full-stack, mechanistic simulator and copilot integration.

### Rakshit Gupta

NSUT · Machine learning engineer, TCN, XGBoost, stacking model, preprocessing and benchmark validation.

MIT LICENSE

SOURCE CODE [github.com/Frisbee148/DigitalTwin](github.com/Frisbee148/DigitalTwin)

Connect signals → estimate state → explain risk → compare scenarios → keep clinicians in control

PATIENT-PORTAL BIOMARKER INGESTION

# Smart acoustic audio and webcam facial distress tracking

Continuous edge pipelines convert spontaneous microphone audio and camera frames into quantified clinical biomarkers and real-time digital twin adaptation.

STREAM 04 • ACOUSTIC SENSING

## Smart Acoustic Cough & Wheeze Sensor

Live microphone audio analyzed via Google YamNet (AudioSet Class 47/50) & Librosa DSP

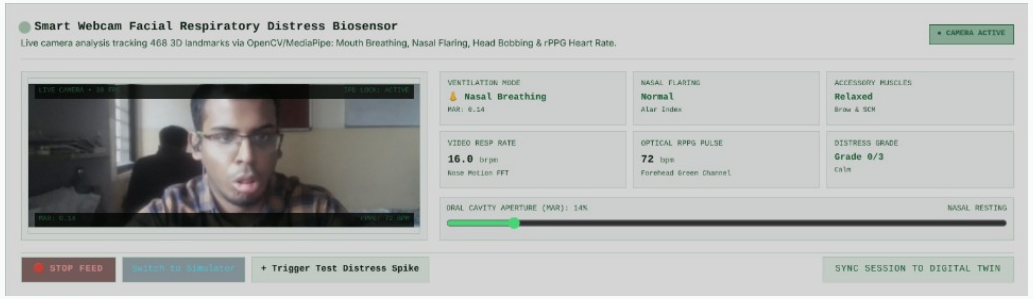



- 512-bin STFT Spectrum: Real-time frequency power & transient cough burst detection
- Tracked Biomarkers: Hourly Cough Count (0 bursts), Graded Severity (0–3), Wheeze (400–1600 Hz)
- Prospective Risk Ingestion: Real-time 48h flare-up risk calculation (62.4% High Risk outlook)

STREAM 05 • FACIAL COMPUTER VISION

## Smart Webcam Facial Distress Biosensor

Live webcam analysis tracking 468 3D landmarks via OpenCV/MediaPipe + optical rPPG




- 468 3D Landmark Mesh: Real-time 30 FPS facial geometry with IPD stabilization lock
- Tracked Biomarkers: Oral Breathing (MAR: 0.14 vs >0.35), Alar Flaring, SCM Accessory Strain
- Contactless Vitals: Forehead green-channel pulse (72 bpm) & nose-motion resp rate (16.0 brpm)

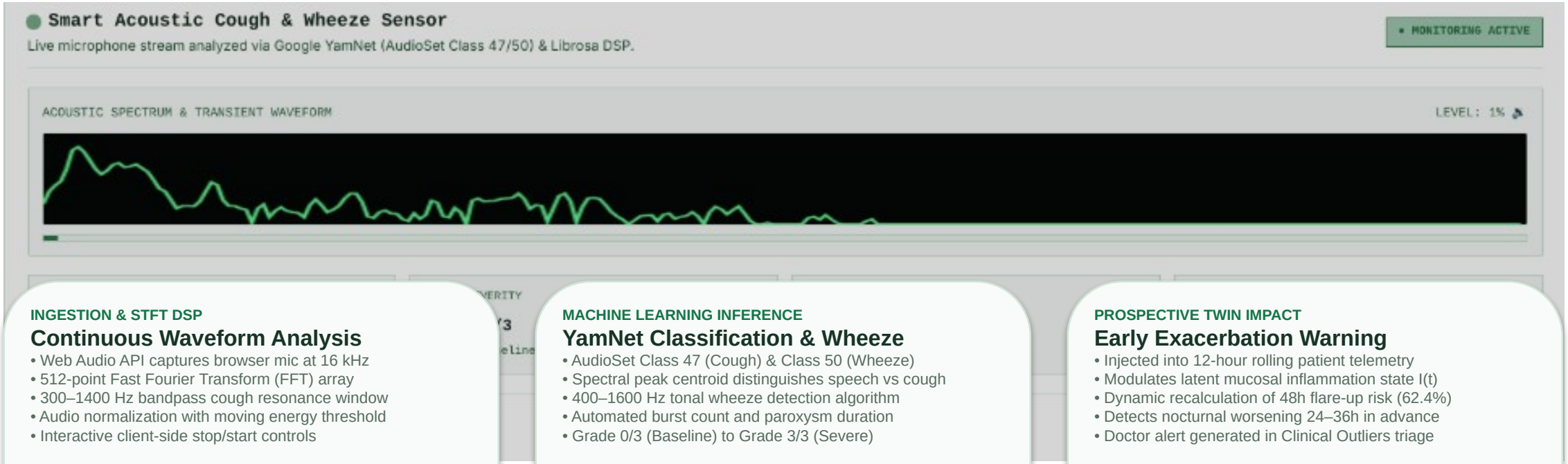
CLOSED-LOOP DIGITAL TWIN INTEGRATION

Acoustic cough bursts and facial respiratory distress telemetry stream into the 12-hour rolling wearable feature window, modulate the latent mucosal inflammation state I(t) in the Euler-Maruyama ODE, and trigger clinician outlier retraining via Elastic Weight Consolidation (EWC).

BIOACOUSTIC SIGNAL PROCESSING

# Acoustic frequency spectrum and transient cough burst detection

Client-side Web Audio API and Google YamNet DSP extract paroxysmal cough frequency and wheeze signatures to detect early mucosal narrowing without hospital spirometry.




INGESTION & STFT DSP

## Continuous Waveform Analysis

- Web Audio API captures browser mic at 16 kHz
- 512-point Fast Fourier Transform (FFT) array
- 300–1400 Hz bandpass cough resonance window
- Audio normalization with moving energy threshold
- Interactive client-side stop/start controls

MACHINE LEARNING INFERENCE

## YamNet Classification & Wheeze

- AudioSet Class 47 (Cough) & Class 50 (Wheeze)
- Spectral peak centroid distinguishes speech vs cough
- 400–1600 Hz tonal wheeze detection algorithm
- Automated burst count and paroxysm duration
- Grade 0/3 (Baseline) to Grade 3/3 (Severe)

PROSPECTIVE TWIN IMPACT

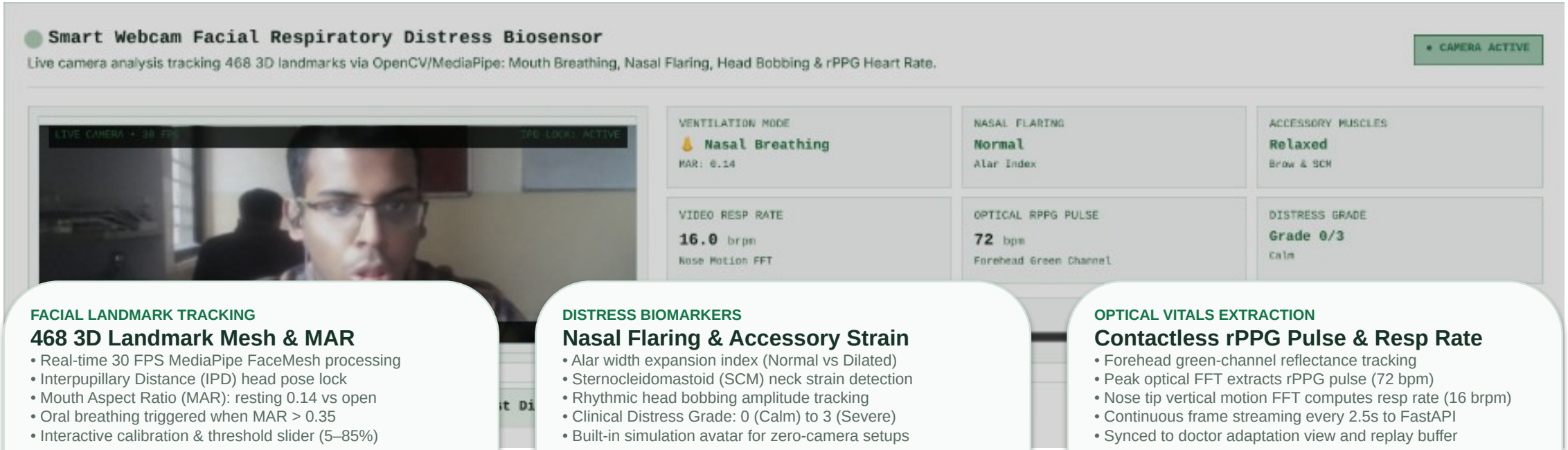
## Early Exacerbation Warning

- Injected into 12-hour rolling patient telemetry
- Modulates latent mucosal inflammation state I(t)
- Dynamic recalculation of 48h flare-up risk (62.4%)
- Detects nocturnal worsening 24–36h in advance
- Doctor alert generated in Clinical Outliers triage

FACIAL BIOMETRICS & OPTICAL DSP

# Optical facial geometry, mouth breathing, and rPPG vital signs

MediaPipe 468 3D landmark mesh and optical green-channel rPPG extract breathing pattern changes, accessory muscle recruitment, and pulse from ordinary webcam video.




FACIAL LANDMARK TRACKING

## 468 3D Landmark Mesh & MAR

- Real-time 30 FPS MediaPipe FaceMesh processing
- Interpupillary Distance (IPD) head pose lock
- Mouth Aspect Ratio (MAR): resting 0.14 vs open
- Oral breathing triggered when MAR > 0.35
- Interactive calibration & threshold slider (5–85%)

DISTRESS BIOMARKERS

## Nasal Flaring & Accessory Strain

- Alar width expansion index (Normal vs Dilated)
- Sternocleidomastoid (SCM) neck strain detection
- Rhythmic head bobbing amplitude tracking
- Clinical Distress Grade: 0 (Calm) to 3 (Severe)
- Built-in simulation avatar for zero-camera setups

OPTICAL VITALS EXTRACTION

## Contactless rPPG Pulse & Resp Rate

- Forehead green-channel reflectance tracking
- Peak optical FFT extracts rPPG pulse (72 bpm)
- Nose tip vertical motion FFT computes resp rate (16 brpm)
- Continuous frame streaming every 2.5s to FastAPI
- Synced to doctor adaptation view and replay buffer